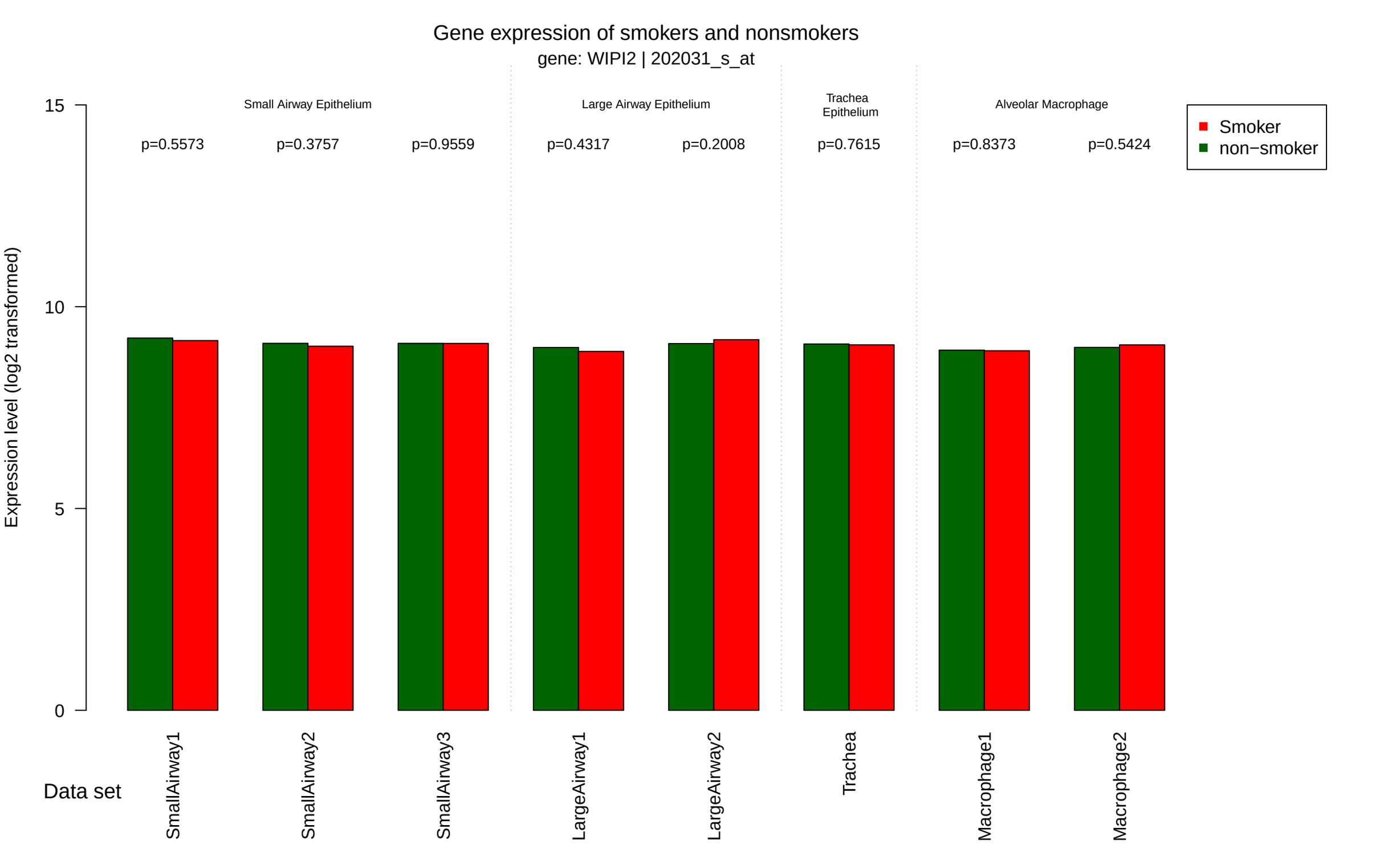
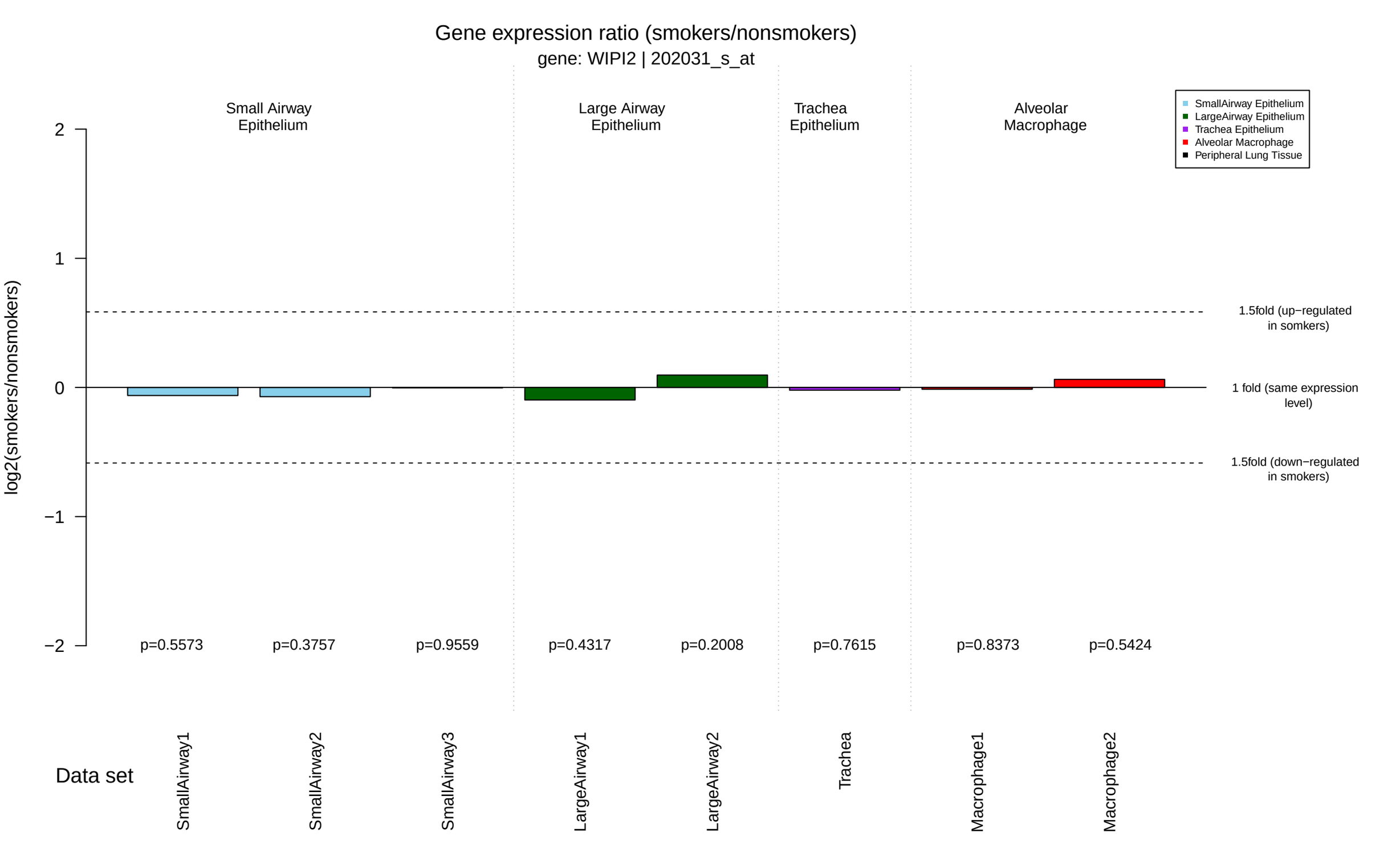
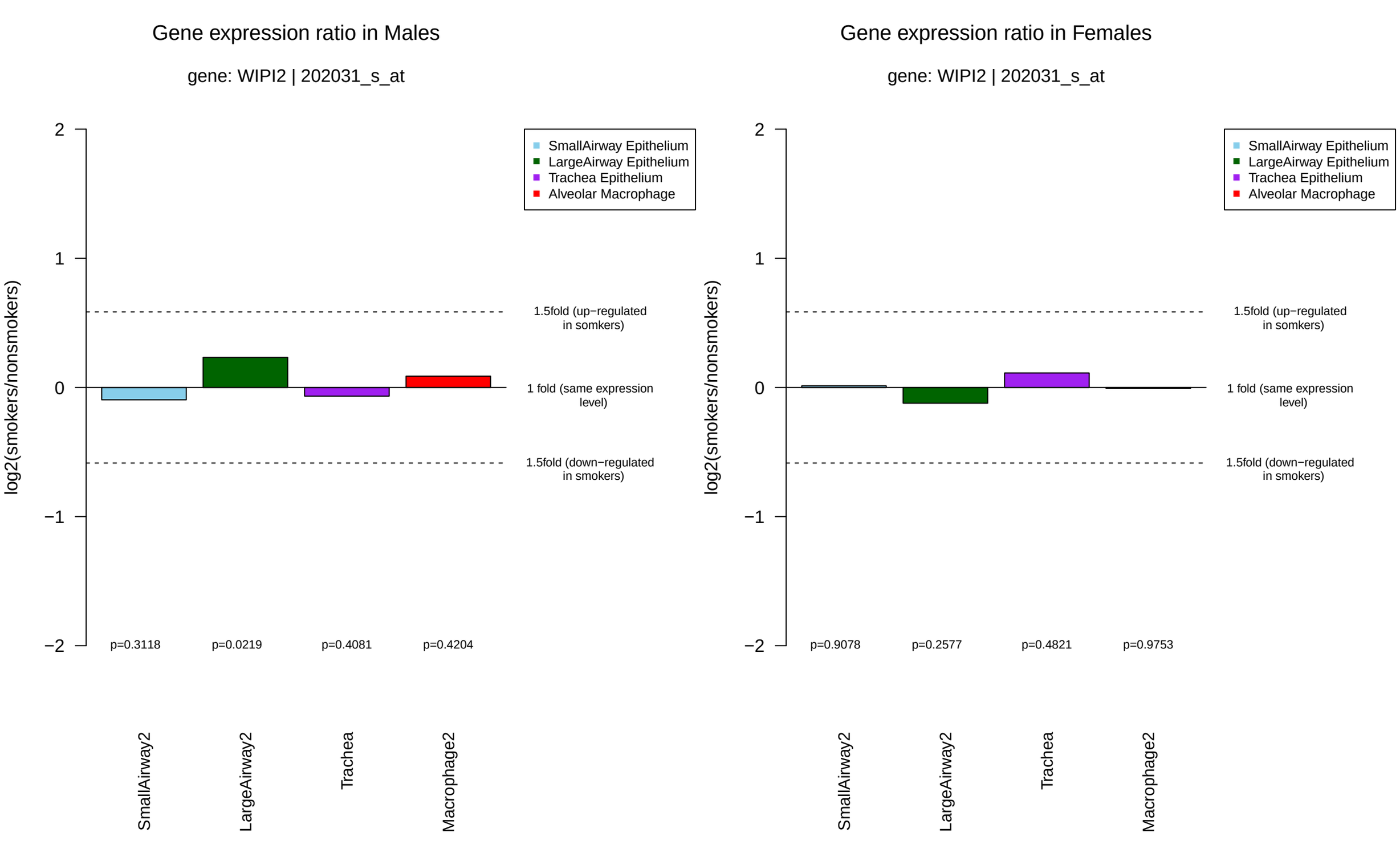
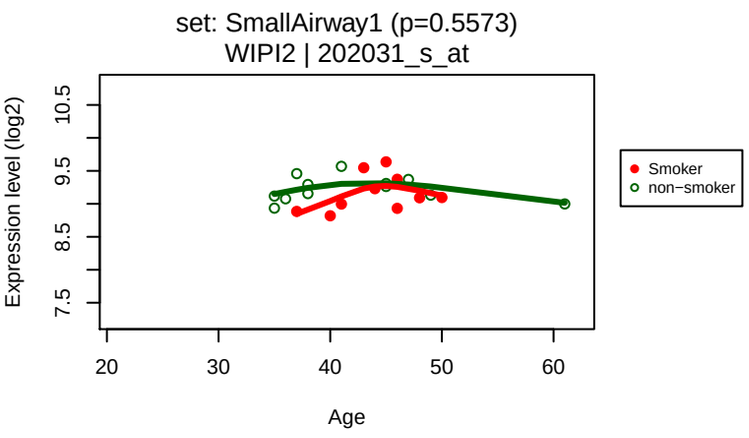
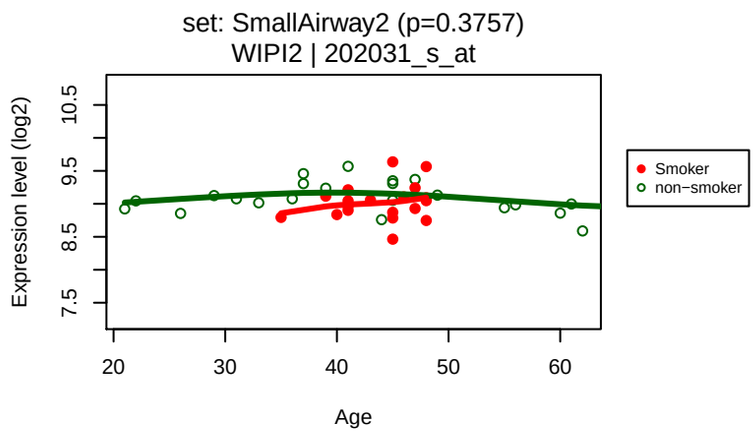
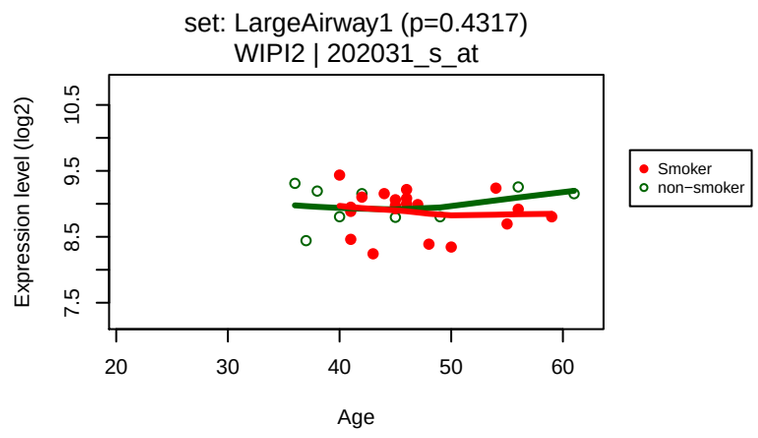

Gene expression of smokers and nonsmokers
gene: WIPI2 | 202031_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5573
p=0.3757
p=0.9559
p=0.4317
p=0.2008
p=0.7615
p=0.8373
p=0.5424
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

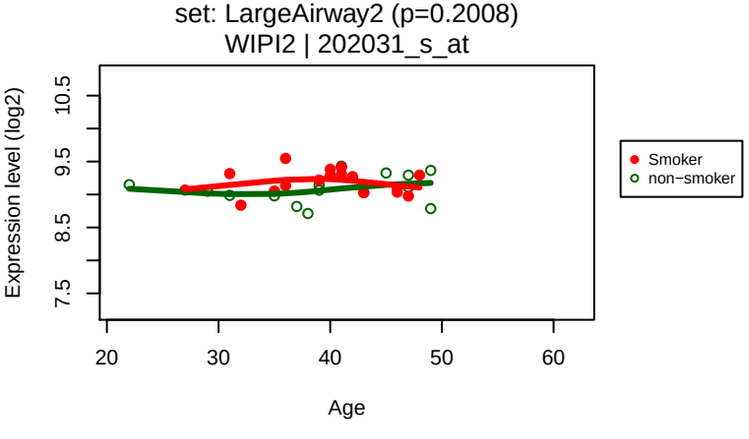

Gene expression ratio (smokers/nonsmokers)
gene: WIPI2 | 202031_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5573
p=0.3757
p=0.9559
p=0.4317
p=0.2008
p=0.7615
p=0.8373
p=0.5424
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

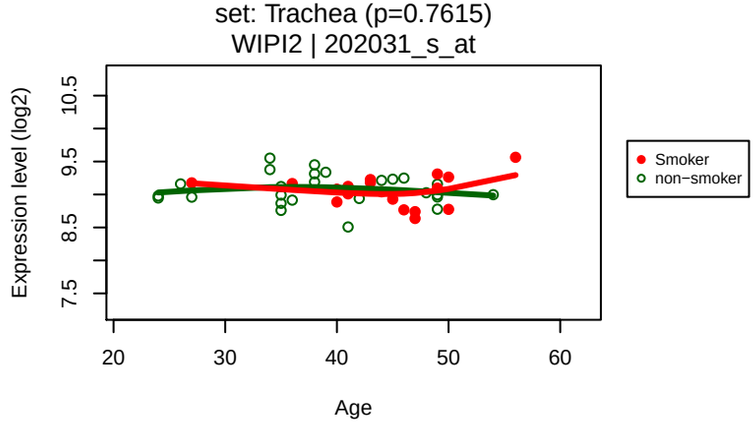

Gene expression ratio in Males
gene: WIPI2 | 202031_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3118
p=0.0219
p=0.4081
p=0.4204
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WIPI2 | 202031_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9078
p=0.2577
p=0.4821
p=0.9753
SmallAirway2
LargeAirway2
Trachea
Macrophage2

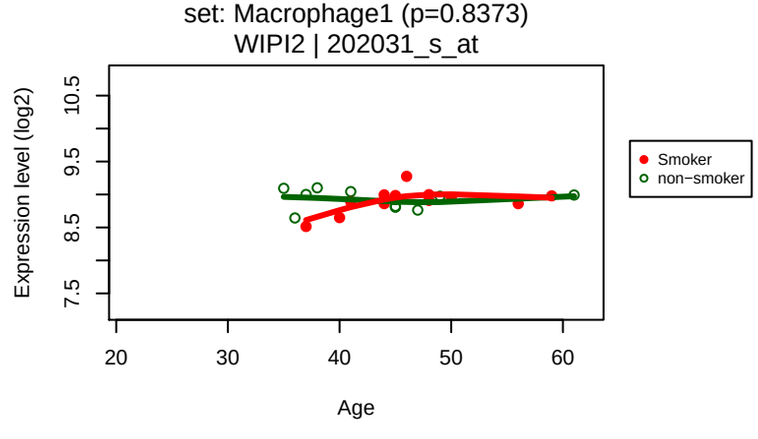

set: SmallAirway1 (p=0.5573)
WIPI2 | 202031_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker

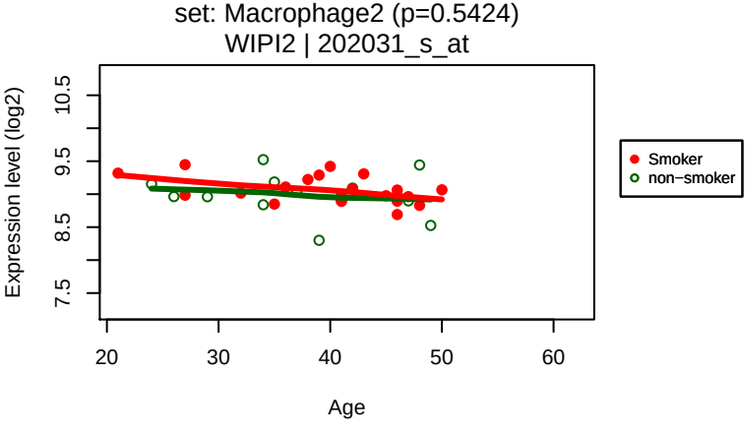

set: SmallAirway2 (p=0.3757)
WIPI2 | 202031_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker

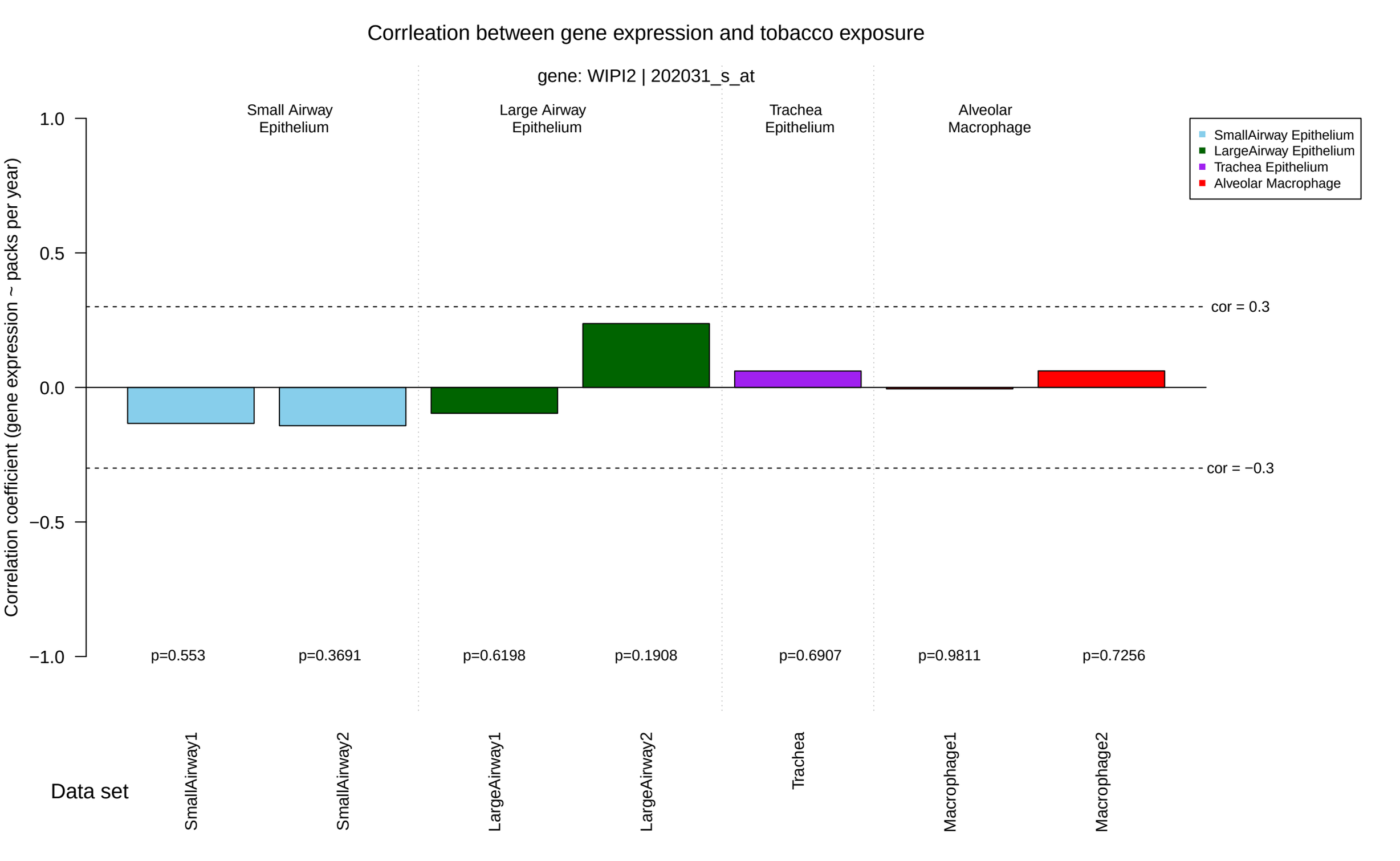

set: LargeAirway1 (p=0.4317)
WIPI2 | 202031_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2008)
WIPI2 | 202031_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.7615)
WIPI2 | 202031_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8373)
WIPI2 | 202031_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5424)
WIPI2 | 202031_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WIPI2 | 202031_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.553
p=0.3691
p=0.6198
p=0.1908
p=0.6907
p=0.9811
p=0.7256
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2